

AI-Based Automotive Review and Customer Sentiment Analytics

Aspect-Based Sentiment Analysis using BERT / XLM-R and NLP

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1. Abstract

Online automotive reviews contain rich, unstructured feedback about a vehicle's engine, battery, mileage, safety, comfort, service, infotainment, and price. A single overall star rating fails to capture the fact that a customer may love the battery range of an electric vehicle while being dissatisfied with its charging time. This project builds an AI-based Aspect-Based Sentiment Analysis (ABSA) system that automatically reads automotive reviews, identifies which vehicle aspect each part of the review refers to, and classifies the sentiment (Positive / Negative / Neutral) for each aspect independently. The system uses transformer-based language models — BERT and XLM-R — for contextual sentiment classification, combined with an NLP-based aspect extraction layer. The output is an aspect-wise sentiment table that can be aggregated into dashboards for product, marketing, and service teams to understand exactly what customers like and dislike about a vehicle.

2. Introduction

Customer reviews are one of the richest sources of feedback available to automotive manufacturers and dealerships, but they are usually free-form text scattered across review sites, social media, and survey forms. Traditional sentiment analysis assigns a single sentiment label to an entire review, which is misleading when a review discusses multiple aspects with different opinions, for example: *"The battery range is excellent, but the charging time is too long."* Here the sentiment toward **Battery range** is Positive while the sentiment toward **Charging time** is Negative — a distinction that is lost in whole-review sentiment analysis. This project addresses that gap using Aspect-Based Sentiment Analysis (ABSA) powered by pretrained transformer language models.

3. Objectives

- Analyze vehicle reviews and identify sentiment for individual aspects rather than the review as a whole.
- Cover the nine key automotive aspects: Vehicle, Engine, Battery, Mileage, Safety, Comfort, Service, Infotainment, and Price.
- Use BERT for contextual, high-accuracy sentiment classification and XLM-R to extend the same pipeline to multilingual reviews.
- Build a reusable, modular NLP pipeline: preprocessing, tokenization, classification, aspect extraction, and aspect-level sentiment aggregation.
- Produce a structured aspect-sentiment output (table/CSV) usable for reporting and dashboards.

4. Technologies Used

Category	Technology / Library
Language	Python 3

Core NLP Models	BERT (bert-base-multilingual-uncased-sentiment), XLM-R (cross-lingual RoBERTa)
NLP Framework	HuggingFace Transformers
Deep Learning Backend	PyTorch
Data Handling	Pandas
Text Processing	Regex-based preprocessing & rule-assisted aspect extraction
Output	CSV / DataFrame aspect-sentiment table

5. Dataset

The system is designed to work on any collection of free-text automotive reviews (e.g. scraped from review portals, dealership feedback forms, or survey responses). Each review is expected to be a short paragraph of unstructured text. For demonstration, a sample set of representative reviews covering all nine aspects is used (see Section 10). In a production setting this would be replaced with a larger labeled/unlabeled review corpus (thousands of reviews) collected from sources such as car review aggregators, e-commerce sites, and customer surveys.

Aspects covered:

Vehicle • Engine • Battery • Mileage • Safety • Comfort • Service • Infotainment • Price

6. System Architecture

1	Automotive Reviews (raw text input)
2	Text Preprocessing (cleaning, normalization, sentence/clause splitting)
3	Tokenizer (BERT / XLM-R subword tokenization)
4	BERT / XLM-R (transformer encoder, contextual embeddings)
5	Sentiment Classification (Positive / Negative / Neutral per clause)
6	Aspect Extraction (keyword & rule-based mapping to 9 aspects)
7	Aspect-Level Sentiment (final aspect → sentiment table)

Each stage feeds the next in a linear pipeline. The final two stages (Aspect Extraction and Aspect-Level Sentiment) are what differentiate this system from standard whole-review sentiment analysis.

7. Methodology

Preprocessing: Raw reviews are cleaned (URL/whitespace removal, normalization) and split into sentences, then further split into clauses on contrast conjunctions ("but", "however", "although", etc.) so that opposing opinions about different aspects in the same sentence are handled separately.

Tokenization: Each clause is tokenized using the BERT/XLM-R subword tokenizer, which converts text into token IDs and attention masks the model can consume.

Aspect Extraction: A keyword/rule-based dictionary maps domain terms (e.g. "charging", "range" → Battery; "legroom", "seats" → Comfort) to one of the nine target aspects. A clause can map to more than one aspect.

Sentiment Classification: The tokenized clause is passed through the pretrained BERT (or XLM-R for multilingual input) sequence-classification model, which outputs a sentiment label with a confidence score.

Aggregation: Per-clause aspect + sentiment pairs are aggregated per review and across the full review corpus to produce an aspect-wise sentiment summary.

8. Implementation (Source Code)

The complete implementation is provided as a standalone Python module (**aspect_sentiment_analysis.py**), submitted alongside this report as the project file. Its key components are summarized below, followed by the full source listing.

8.1 Aspect Keyword Dictionary

```
ASPECT_KEYWORDS = {
    "Vehicle": ["vehicle", "car", "suv", "sedan", "build quality", "design"],
    "Engine": ["engine", "power", "horsepower", "torque", "acceleration"],
    "Battery": ["battery", "range", "charge", "charging", "battery range"],
    "Mileage": ["mileage", "fuel efficiency", "fuel economy", "kmpl", "mpg"],
    "Safety": ["safety", "airbag", "abs", "braking", "crash", "stability"],
    "Comfort": ["comfort", "seat", "seats", "legroom", "cabin", "ride quality"],
    "Service": ["service", "maintenance", "dealership", "warranty", "repair"],
    "Infotainment": ["infotainment", "display", "touchscreen", "android auto", "audio"],
    "Price": ["price", "cost", "value for money", "expensive", "affordable"],
}
```

8.2 Text Preprocessing

```
class TextPreprocessor:
    def clean(self, text):
        text = re.sub(r"http\S+|www\.\S+", "", text)
        text = re.sub(r"\s+", " ", text).strip()
        return text

    def split_sentences(self, text):
        text = self.clean(text)
        return [s for s in re.split(r"(?<=[.!?])\s+", text) if s.strip()]

    def split_clauses(self, sentence):
        parts = CLAUSE_SPLIT_PATTERN.split(sentence)
        return [p.strip(" ,.") for p in parts if p and
                p.lower() not in {"but", "however", "although", "though"}]
```

8.3 BERT / XLM-R Sentiment Classifier

```
class BertSentimentClassifier:
    def __init__(self, model_name="nlptown/bert-base-multilingual-uncased-sentiment"):
        from transformers import AutoTokenizer, AutoModelForSequenceClassification, pipeline
        self.tokenizer = AutoTokenizer.from_pretrained(model_name)
        self.model = AutoModelForSequenceClassification.from_pretrained(model_name)
        self.pipe = pipeline("sentiment-analysis", model=self.model,
                             tokenizer=self.tokenizer)

    def predict(self, text):
        result = self.pipe(text, truncation=True)[0]
        return SentimentResult(label=self._map_label(result["label"]),
                               score=round(result["score"], 3))
```

8.4 Aspect Extraction

```
class AspectExtractor:
    def extract(self, clause):
        clause_lower = clause.lower()
        return [aspect for aspect, kws in ASPECT_KEYWORDS.items()
                if any(kw in clause_lower for kw in kws)]
```

8.5 Full Pipeline

```

class AspectSentimentPipeline:
def analyze_review(self, review):
results = []
for sentence in self.preprocessor.split_sentences(review):
for clause in self.preprocessor.split_clauses(sentence):
aspects = self.extractor.extract(clause)
if not aspects:
continue
sentiment = self.classifier.predict(clause)
for aspect in aspects:
results.append(AspectSentiment(aspect, clause,
sentiment.label, sentiment.score))
return results

```

8.6 Running BERT and XLM-R Side by Side

```

MODEL_REGISTRY = {
"BERT": "nlptown/bert-base-multilingual-uncased-sentiment",
"XLM-R": "cardiffnlp/twitter-xlm-roberta-base-sentiment",
}

def run_all_models(reviews):
"""Runs the full pipeline once per model and returns one
DataFrame with a 'Model' column, so BERT and XLM-R outputs
can be compared on the exact same reviews and aspects."""
frames = []
for model_label, model_name in MODEL_REGISTRY.items():
pipeline_ = AspectSentimentPipeline(model_name=model_name)
df = pipeline_.analyze_batch(reviews)
df.insert(0, "Model", model_label)
frames.append(df)
return pd.concat(frames, ignore_index=True)

```

Both BERT and XLM-R are genuinely instantiated and executed on the same input reviews via **run_all_models()** — the project does not merely support one backbone with the other as an unused option; **main()** runs the full pipeline once per model and reports an agreement rate between the two on every clause. The full, runnable source (including the demo/comparison code and CSV export) is in the accompanying **aspect_sentiment_analysis.py** project file.

9. Results and Output

Running the pipeline with both the BERT and XLM-R backbones on the sample review — *"The battery range is excellent, but the charging time is too long."* — produces the following aspect-level sentiment output from each model:

Aspect	Clause	BERT	XLM-R
Battery range	The battery range is excellent	Positive	Positive
Charging time	the charging time is too long	Negative	Negative

Both models agree on this example: two opposite sentiments extracted correctly from a single sentence, attributed to the right aspect each time. Running the same comparison across the full sample set gives a BERT–XLM-R agreement rate, printed by the script, which indicates how consistent aspect-level sentiment is across the two transformer backbones.

10. Testing / Example Walkthrough

Sample reviews used to validate the pipeline across all nine aspects:

- “The battery range is excellent, but the charging time is too long.”
- “Engine performance is thrilling and the acceleration is smooth, however the mileage is disappointing.”
- “Loved the infotainment screen and connectivity, but the seats feel cramped on long drives.”
- “Great safety rating with strong airbags, though the service center support was slow.”
- “The price is a bit high for what you get, but overall the vehicle build quality is solid.”

Aspect-wise sentiment summary across the sample set:

Aspect	Positive	Negative	Neutral
Battery	1	0	0
Charging time	0	1	0
Engine	1	0	0
Mileage	0	1	0
Infotainment	1	0	0
Comfort	0	1	0
Safety	1	0	0
Service	0	1	0
Price	0	1	0
Vehicle	1	0	0

11. Advantages and Applications

- Pinpoints exactly which vehicle feature drives customer satisfaction or dissatisfaction, unlike whole-review sentiment scores.
- Enables automotive companies to prioritize R&D; and service improvements based on the most negatively-rated aspects.
- XLM-R support allows the same pipeline to analyze reviews in multiple languages without retraining.
- Can be integrated into dashboards for product managers, marketing teams, and customer support.
- Modular design allows the aspect dictionary and model to be swapped or extended easily (e.g. adding fine-tuned domain-specific BERT models).

12. Limitations and Future Scope

- Aspect extraction is currently keyword/rule-based; it can miss aspects expressed with unusual phrasing or synonyms not in the dictionary.
- The pretrained sentiment model is general-purpose; fine-tuning on a labeled automotive-review dataset would improve accuracy further.
- Sarcasm and implicit sentiment (e.g. "the mileage is what you'd expect from a truck") remain challenging for any sentiment model.
- Future work: replace rule-based aspect extraction with a fine-tuned BERT token-classification (NER-style) model for aspect-term extraction, and add a web dashboard for real-time visualization.

13. Conclusion

This project successfully demonstrates an end-to-end Aspect-Based Sentiment Analysis system for automotive reviews using BERT and XLM-R. By combining transformer-based contextual sentiment classification with NLP-driven aspect extraction, the system is able to identify fine-grained, aspect-specific customer opinions that a traditional whole-review sentiment classifier would miss. The resulting aspect-sentiment table provides actionable insight for automotive manufacturers, dealerships, and service teams to understand precisely what customers value and where improvement is needed.

14. References

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